

Repair Manuals

Timing chain: removal/installation

Warnings and recommendations

Unless otherwise advised by the manufacturer, the following procedures are recommended:

Always renew the timing chain

If the timing chain is to be re-used, mark the running direction before removal

Never turn the engine in the direction opposite to that of normal operation

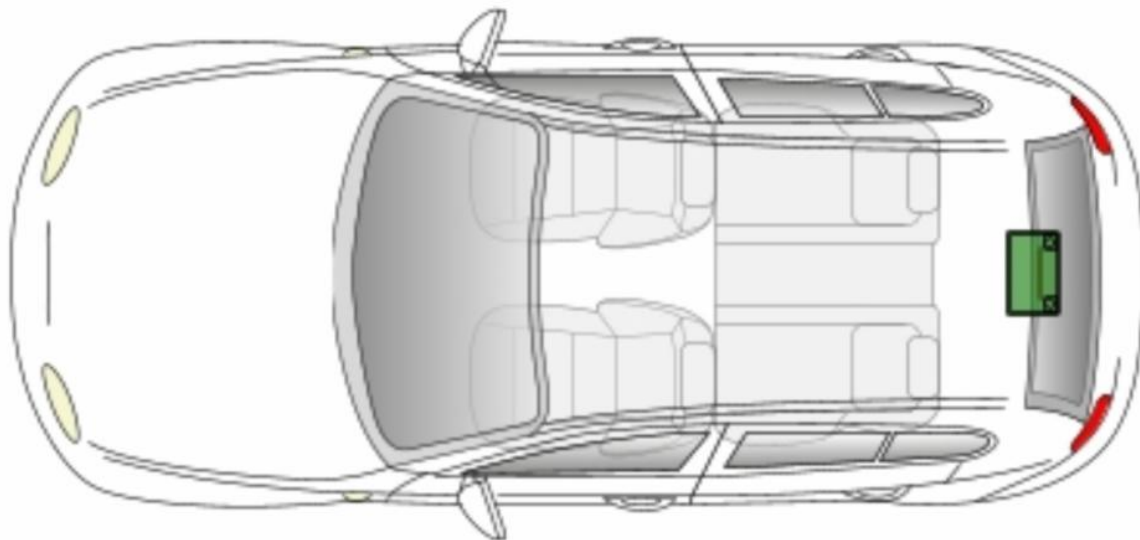
Note: Avoid bending, torsion or traction

Always use new gaskets

General

Before disconnecting the battery cable, check the audio system security code

Battery location



Luggage compartment

Clean off the old sealant

Apply sealant

Renew the seals

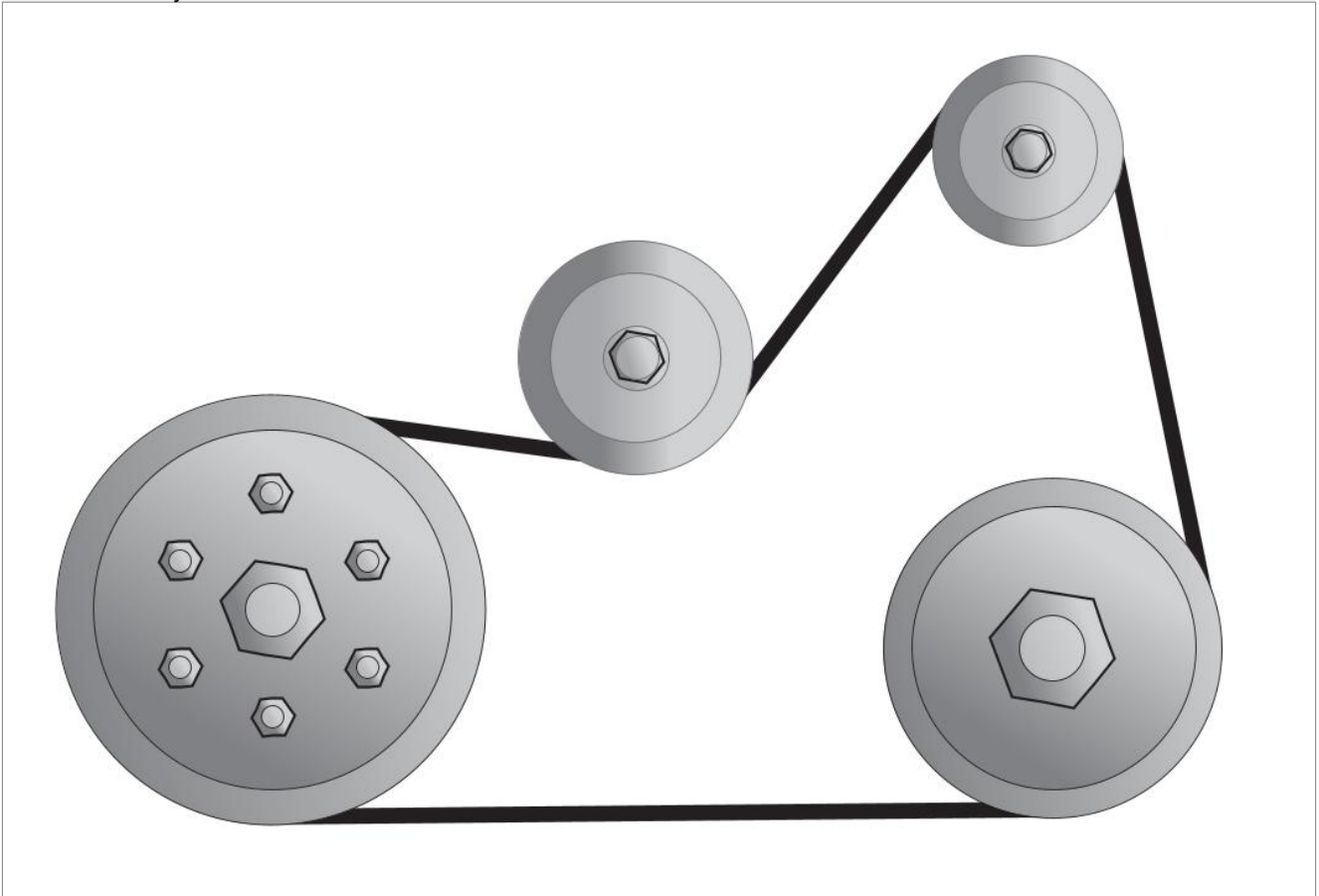
Removal

Disconnect the battery

Repair Manuals

Remove the inlet silencer
Remove the bulkhead cover
Remove the bulkhead cover insulation
Remove the wiring loom from the cylinder head
Remove the valve cover
Remove the spark plugs
Remove the coolant pump friction wheel
Remove the ancillary drive belt

Drive belt layout

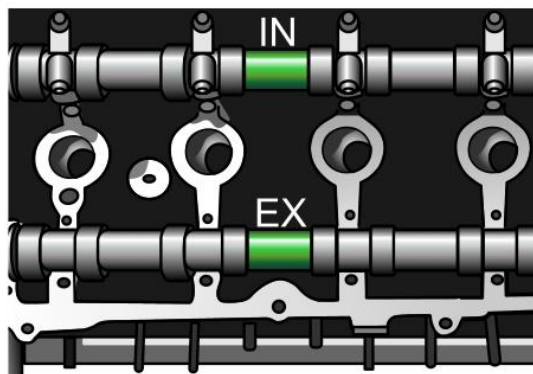


Remove the crankshaft pulley
Turn the crankshaft clockwise

Vehicles with automatic transmission:
Use a special tool to lock the crankshaft (2 299 362)

Vehicles with manual gearbox:
Use a special tool to lock the crankshaft (0 496 709)

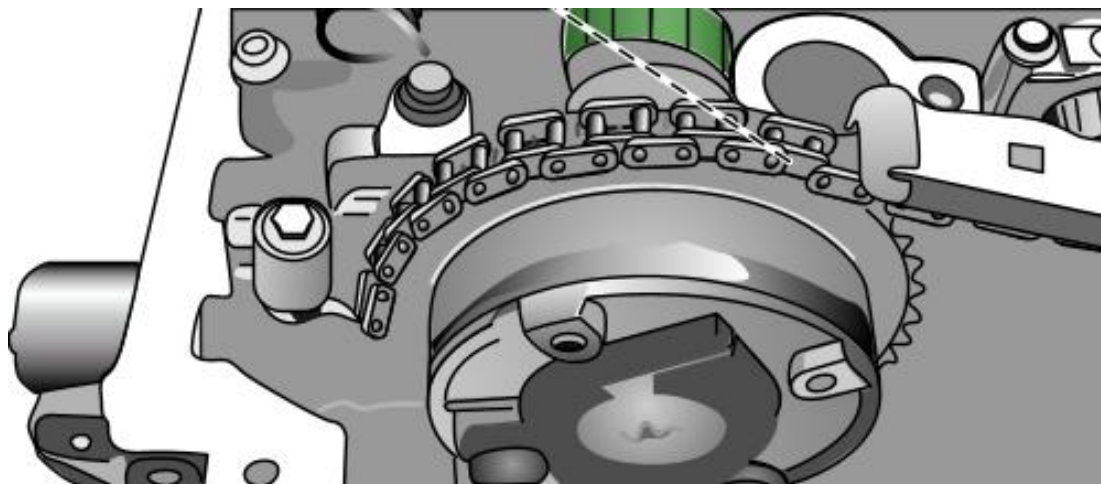
All vehicles:
Variable valve timing actuator:
Turn the camshafts to check that the camshaft adjustment units can be locked
Check the camshaft timing marks (IN, EX)

Repair Manuals

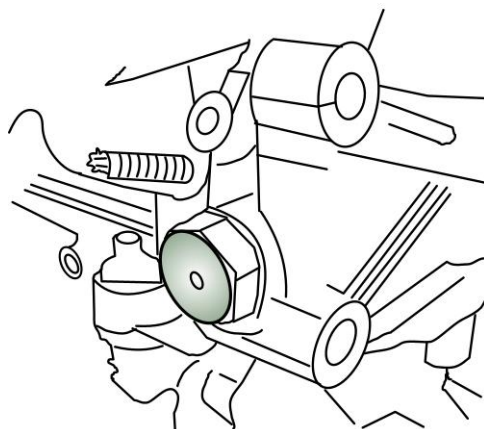
Exhaust camshaft cam angle



Inlet camshaft cam angle

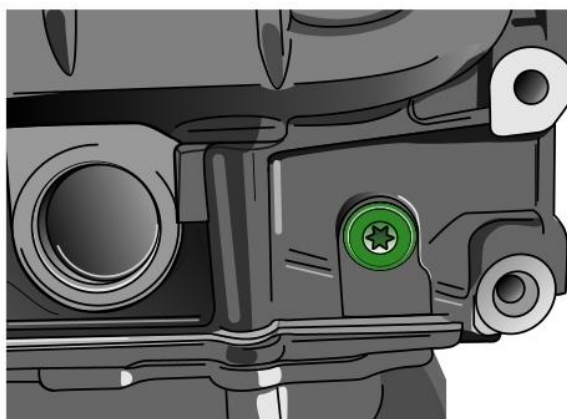


Note: Both camshafts must point upwards at an angle
Use the special tools to lock the camshafts (11 7 440)
Remove the chain tensioner piston

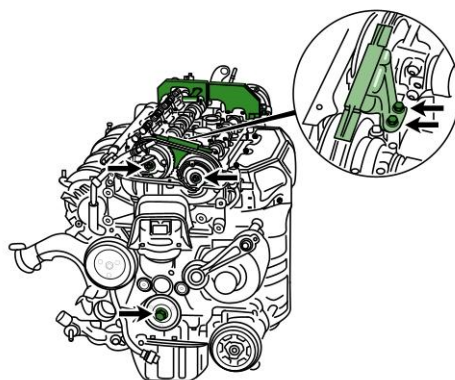
Repair Manuals

Chain guide rail:

Remove the bolt

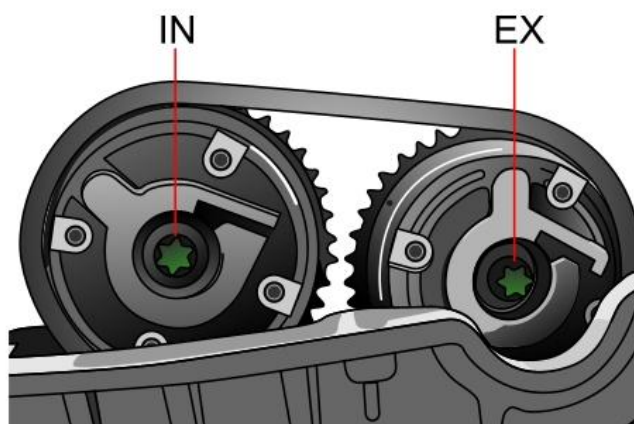


Remove the timing chain upper guide



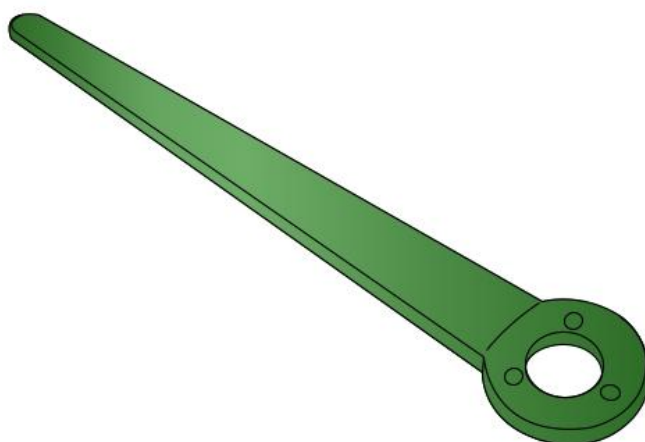
Remove the inlet camshaft timing adjuster (IN)

Remove the exhaust camshaft timing adjuster (EX)

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Remove the crankshaft gearwheel centre bolt

Use the special tool to counterhold (11 9 280)



Bearing pins for guide and tensioner rails:

Remove the bolts



Remove the crankshaft hub

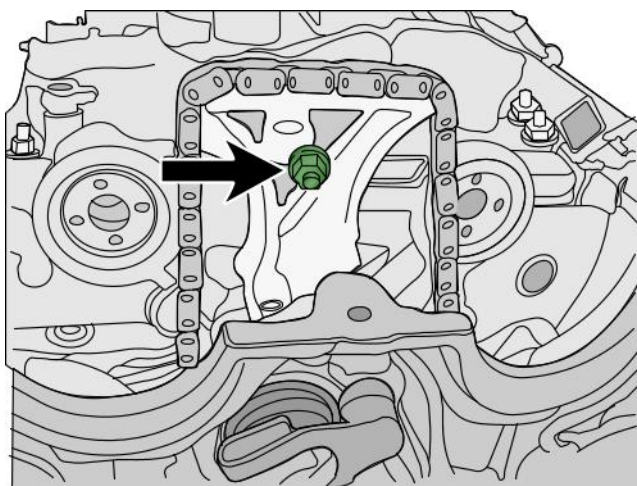
Remove the chain and the crankshaft gearwheel

Remove the chain guide with the timing chain, guide rail and tensioner rail

Note: Check that the oil pump gearwheel and chain are in place

Remove the timing chain guide bolts from the cylinder head (x 2)

Repair Manuals



Remove the chain guide with the timing chain, guide rail and tensioner rail

Installation

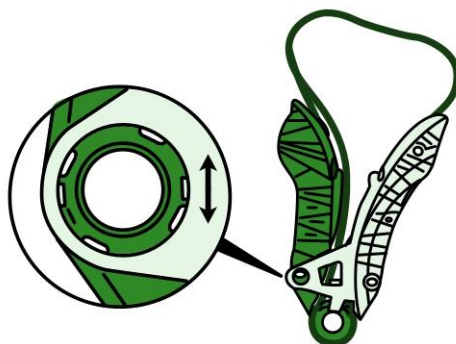
Timing chain assembly:

Fit the chain tensioner guide

Fit the chain guide

Fit the timing chain

Fit the crankshaft gearwheel to the timing chain



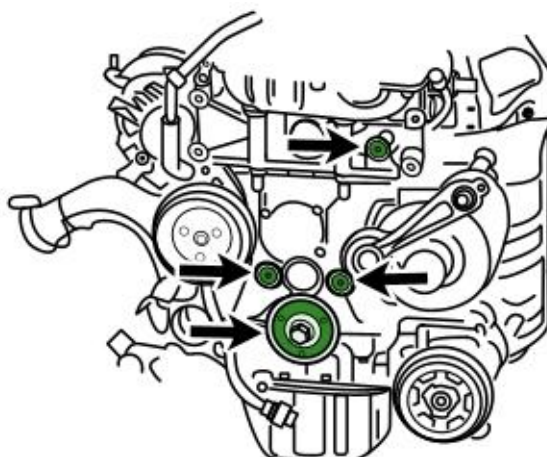
Fit the timing chain assembly from above the cylinder head

Note: Check that the oil pump gearwheel and chain are in place

Refit the timing chain rail bolts

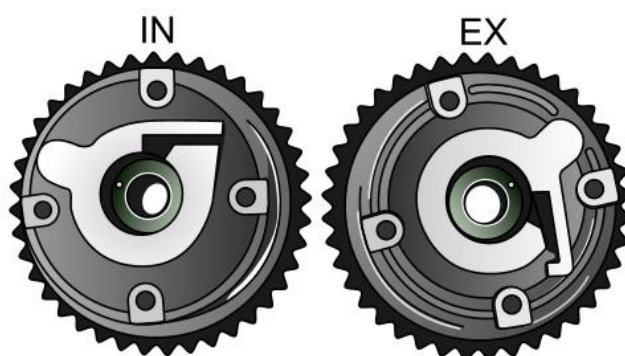
Refit the crankshaft hub

Repair Manuals



VANOS unit:

Check the marks (IN, EX)



Refit the exhaust camshaft timing adjuster (EX)

Do not fully tighten

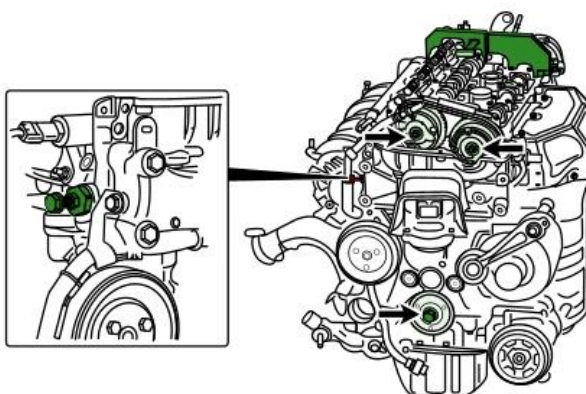
Refit the inlet camshaft timing adjuster (IN)

Do not fully tighten

Refit the timing chain upper guide

Fit the dummy chain tensioner (11 9 340)

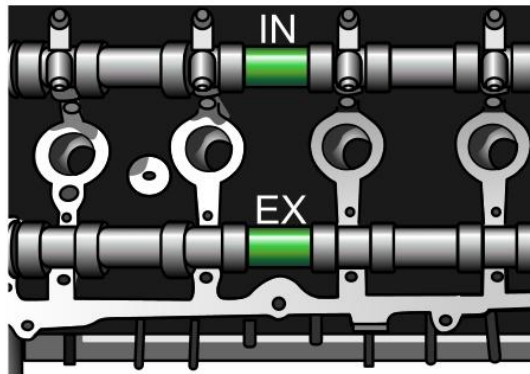
Pre-tension the timing chain (0.6 Nm)



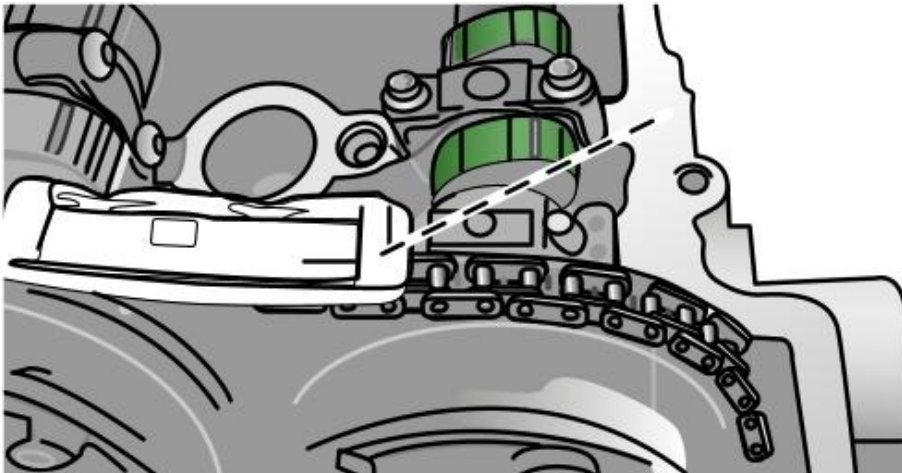
Repair Manuals

Tighten the inlet camshaft gearwheel
Tighten the exhaust camshaft gearwheel
Remove the dummy chain tensioner (11 9 340)
Refit the chain tensioner piston

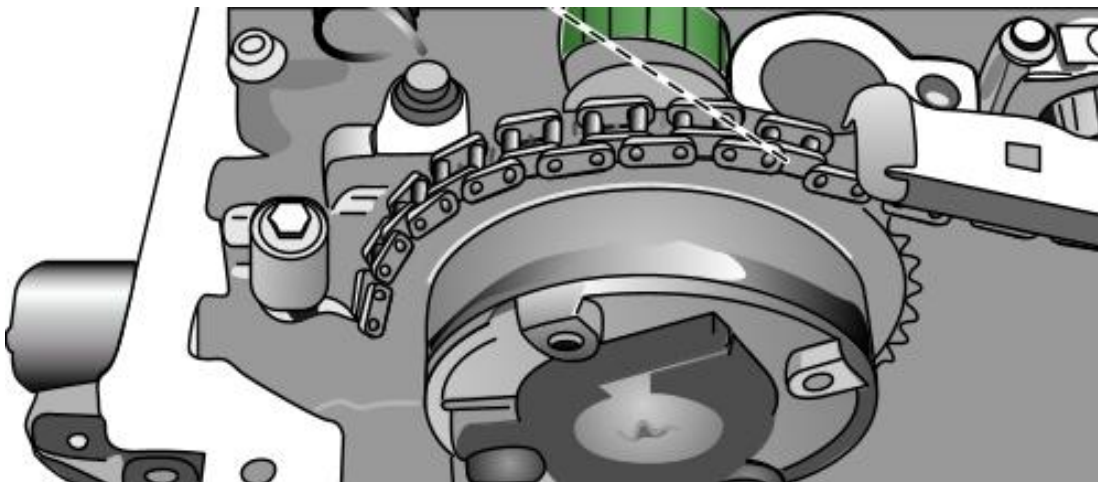
Check the camshaft timing marks (IN, EX)



Exhaust camshaft cam angle



Inlet camshaft cam angle



Note: Both camshafts must point upwards at an angle

Repair Manuals

Further refitting is in the reverse order of removal

Torque settings

Valve cover

Valve cover 10 (Nm)

Camshaft gearwheel

Camshaft gearwheel

Renew the bolts

Stage 1 20 (Nm)

Stage 2 180 (°)

Timing chain tensioner

Timing chain tensioner (Upper section) 80 (Nm)

Renew the seal

Crankshaft pulley

Crankshaft pulley (Outer bolts) 28 (Nm)

Crankshaft gearwheel

Crankshaft gearwheel (Centre bolt)

Renew the bolt (Threads lightly oiled)

Stage 1 50 (Nm)

Stage 2 180 (°)

Timing chain guide

Timing chain guide (Upper) 25 (Nm)

Renew the seal

Timing chain guide (Lower) 24 (Nm)

Renew the seal

Spark plugs

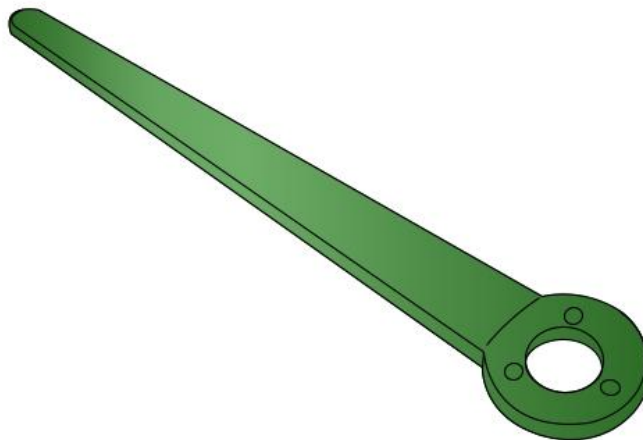
Spark plugs (M12) 20 - 26 (Nm)

Spark plugs (M14) 27 - 33 (Nm)

Coolant pump friction wheel (M6: 10 Nm)

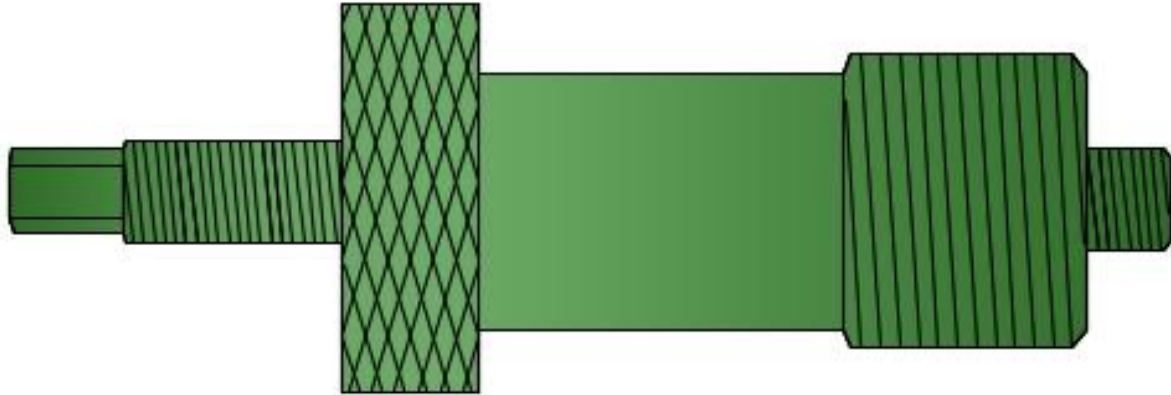
Special tools

Crankshaft centering tool: (11 9 280)

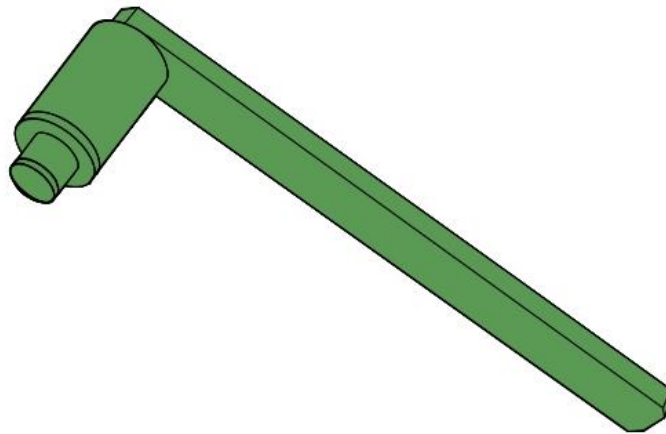


Adaptor: (11 9 340)

Repair Manuals



Flywheel locking tool: (0 496 709)



Flywheel locking tool: (2 299 362)

